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Online Health Consultation Portal

Design Document

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REVIEW 1

I. Acknowledgement

First, we would like to express our gratitude to our dedicated instructor – Mr. Hoang Duc Quang. We would have not been able to finish this dissertation without his guidance and encouragement. During the time of the project, he continuously gave us constructive advises and showed his passion in teaching to help us reach the expected result of the projects. Therefore, we truly value his dedication and we learned a lot from him.

Next, we fully appreciate our team members who worked day and night to carry out and complete this project. Although sometimes there were challenges, but by through them, we learned a lot and our team bonds became stronger. The project is the result of our relentless efforts to improve through every step, our strong commitment to the expected outcome and our passionate spirit at one team.

Finally, we would send a huge thanks to Aptech's eProject team who gave us an excellent chance to apply what we have learnt theoretically into real products and gain experience as well as innovative mindset which is extremely vital for our future career and our end users.

II. Problem Definition

1. Introduction:

Access to healthcare professionals can be difficult, especially for people living in remote areas or for those with limited mobility. Traditional healthcare systems may involve long wait times for appointments, geographical constraints, and inadequate access to specialized care. Moreover, maintaining a patient's health history across various visits and institutions can often lead to fragmented data, which hinders efficient treatment. A comprehensive platform that allows for easy scheduling of virtual consultations, seamless management of health records, and basic AI-driven diagnoses could significantly improve access to healthcare, reduce wait times, and streamline the entire healthcare process.

The Online Health Consultation Portal aims to provide an efficient, secure, and accessible way for patients to receive medical consultations, store health records, and use AI for preliminary diagnosis. By reducing barriers to access and providing a more streamlined healthcare experience, this platform has the potential to revolutionize the way people engage with healthcare services.

2. Existing System

The Online Health Consultation Portal is a secure web-based platform that allows patients to book virtual consultations with doctors, maintain and access their health records, and receive basic AI-driven diagnostics. This solution aims to enhance access to healthcare, especially for those who are unable to visit a healthcare facility physically. By providing tools for managing health records digitally and utilizing AI for preliminary diagnoses, the platform will make healthcare more efficient, accessible, and patient-centric.

III. Requirement Specification

1. Patient Requirements

1.1. Personal Account Management:

- The system shall allow patients to register a new account by providing personal and health-related details. A new record shall be created in the Users, Patients, and HealthRecords tables.
- The system shall support secure login and logout operations with proper authentication and session handling.

- The system shall allow patients to update personal information, including FullName, DateOfBirth, and insurance data stored in the Patients table.

1.2. Appointment Management:

- The system shall allow patients to book new appointments by selecting a doctor, consultation type, and appointment time. A new record shall be created in the Appointments table with Status = "Scheduled".
- The system shall display upcoming and past appointments retrieved from the Appointments table.
- The system shall allow patients to cancel appointments by updating the Status to "Cancelled", provided that the cancellation occurs at least 24 hours before the scheduled time.

1.3. Health Record Management:

- The system shall display a summary of the patient's medical history (MedicalHistorySummary) from the Patients table.
- The system shall allow patients to upload new medical documents, generating linked records under their HealthRecordID.
- The system shall allow patients to view and download previously uploaded medical documents.

1.4. Post-Consultation Interaction:

- The system shall allow patients to view prescriptions (Prescriptions) issued after consultations.
- The system shall allow patients to submit reviews and ratings for doctors by creating a new record in the Reviews table once the appointment Status = "Completed".
- The system shall allow patients to view invoices (Invoices) generated for completed consultations.

2. Doctor Requirements.

2.1. Doctor Profile Management:

- The system shall allow doctors to log in and access their profile page.
- Doctors shall be able to update their professional information, including Qualifications, Specialty, YearsOfExperience, and Location in the Doctors table.

- The system shall store and maintain all profile updates.

2.2. Appointment and Work Schedule Management:

- The system shall allow doctors to define their weekly consultation availability.
- The system shall display a dashboard view of all appointments by date, week, and status (Scheduled).
- The system shall allow doctors to confirm or cancel appointments, updating the Status accordingly and notifying patients automatically.

2.3. Consultation Process:

- The system shall allow doctors to access patient health records and medical documents during consultations.
- Doctors shall be able to record diagnoses, treatment plans, and notes in the DoctorNotes section.
- Upon completion, the system shall create a record in the Consultations table and update the corresponding appointment Status to "Completed".

2.4 Prescription Management:

- The system shall allow doctors to create and issue electronic prescriptions for patients after consultations.
- Doctors shall provide information including MedicationName, Dosage, and Instructions.
- The system shall generate prescription records in the Prescriptions table linked to the ConsultationID and send notifications to patients.

3. Administrator Requirements:

3.1 User Management:

- The system shall allow administrators to log in to the admin control panel.
- Administrators shall be able to view all registered users from the Users table along with details from the Patients and Doctors tables.
- Administrators shall be able to:
- Create new doctor accounts after verification.

- Deactivate user accounts violating platform policies by updating the IsActive field.
- Update user information stored in the system.

3.2 Admin Dashboard for Healthcare Providers:

- The system shall provide an administrative dashboard for healthcare providers to:
 - Manage consultations and track patient progress.
 - Access and utilize diagnostic tools.
- The system shall allow administrators to review, approve, and update patient health records as part of the oversight process.

4. Functional Requirement Specifications

2.1. Function of User:

2.1.1. Un-Registered Users:

REQ-01. Register with the portal after making a payment.

REQ-02. Schedule virtual consultations with doctors (general practitioners, specialists) based on availability and preferred time.

REQ-03. View doctor profiles including qualifications, specialties, years of experience, and patient ratings/reviews.

REQ-04. Access a secure, cloud-based platform to store and manage health records (medical history, prescriptions, lab results, imaging reports, past consultations).

REQ-05. Upload documents and images (e.g., scans, X-rays) for doctor review.

REQ-06. Send feedback to the portal.

REQ-07. View health reminders and notifications (e.g., consultations, medication refills, vaccinations, follow-ups).

2.1.2 Registered Users:

REQ-08. All functions available to un-registered users.

REQ-09. Login and logout from the website.

REQ-10. View and manage their health records, including treatment progress over time.

REQ-11. Reschedule or cancel appointments with prior notice.

REQ-12. Participate in video consultations with screen sharing for diagnoses and treatment plans.

REQ-13. Receive prescriptions and follow-up appointment reminders via the platform.

REQ-14. Use secure chat/text-based communication for questions and asynchronous messaging.

2.2 Function of Admin:

REQ-15. Login, logout, and change password.

REQ-16. Manage virtual consultation scheduling and doctor profiles via an admin dashboard.

REQ-17. Review, approve, and update patient health records.

REQ-18. View and manage orders (e.g., consultation invoices/bills).

REQ-19. Manage user accounts (view, change password, delete).

REQ-20. View and delete feedback.

REQ-21. Manage health reminders and notifications for patients.

5. Non-Functional Requirement Specifications:

REQ-21. Good User Interface, User Experience.

REQ-22. Clean code, project.

REQ-23. Try hard improve the performance.

REQ-24. Try this website run on all browsers.

REQ-25. Responsive this website on mobile, tablet, desktop.

REQ-26. Try limit bugs on this website.

6. System requirement specifications

4.1. Server requirements:

❖ Hardware:

Component	Requirement
CPU	Processor type: Dual-core processor (Intel Core i3 or equivalent) or faster Processor speed: Recommended: Minimum: 2.0 GHz
OS	Windows Server 2019 or later with IIS, or Linux (Ubuntu 20.04 LTS)
Memory (RAM)	RAM: Minimum: 2 GB Recommended: 4 GB or more
Hard Drive	Free space: Minimum: 20 GB SSD

❖ Software:

Component	Requirement
ASP.NET Core	Version .NET 8
RDBMS	SQL Server 2019 or later

4.2. Client Requirement

❖ Hardware:

Component	Requirement
CPU	Processor type: Single-core processor (Intel Pentium or equivalent) or faster Processor speed: Recommended: 1.5 GHz or faster
OS	Windows 10, macOS Monterey or later, Android 11+, iOS 15+

Memory (RAM)	RAM: Minimum: 2 GB
Hard Drive	Free space: Minimum: 10 GB

❖ Software:

Component	Requirement
Web Browser	Latest versions of Google Chrome 120+, Mozilla Firefox 120+, Microsoft Edge 120+, Safari 16+

4.3. Development Software

- ❖ SQL Server.
- ❖ Microsoft Visual Studio 2022.

4.4. Technology

- ❖ ASP.Net core 9.0.10
- ❖ Entity Framework Core 9.0.10

TASK SHEET REVIEW 1

Project Ref. No.:		Project Title:	Activity Plan Prepared By:	Date of Preparation of Activity Plan:			
eP/Advertisement Portal Management System/01							
St. No.	Task			Actual Start Date	Actual Days	Team Mate Names	Status
1	Acknowledgement	Online Health Consultation Portal	Do Thanh Hung	09/11/2025	1	Dat	Completed
2	Problem Definition			09/11/2025	1	Huy	Completed
3	Requirement Specification			09/11/2025	1	Hiep	Completed
4	Task sheet review			09/11/2025	1	Hung	Completed
Date: 09/11/2025							
Signature of Instructor:				Signature of Team Leader:			
Mr. Hoang Duc Quang				Do Thanh Hung			